1. Clinical Question & Core Hypothesis

Clinical Question

Why do some tumors with favorable immune architecture still fail to achieve better clinical outcomes?

Working Hypothesis

Main hypothesis: The durability of intravesical BCG–induced tumor control in NMIBC depends on a three-way balance among (a) organized local immunity represented by TLS-like lymphoid aggregates (and quantified by Immunoscore), (b) tumor immune contexture defined by PD-L1 expression (CD274) and tumor mutational burden, and (c) genomic instability factors (MSI status, TP53 and ER context) that shape neoantigen availability. TLS presence is necessary for effective local priming but is not sufficient unless the tumor milieu permits sustained effector activity; high PD-L1 expression suppresses TLS-driven responses unless neoantigen load is high enough to overcome this suppression. Consequently, the strongest and most durable responses to intravesical BCG will occur in TLS-positive tumors with low PD-L1 and high neoantigen load (high bTMB/MSI or favorable TP53/ER contexts). Conversely, TLS-positive tumors with high PD-L1 and low neoantigen load will more often recur despite apparent local immune organization. Noninvasive biomarkers that reflect these tissue-level interactions (urine DNA methylation signatures and ADNP expression in CTCs) will correlate with TLS status, neoantigen load, and ultimate therapy response, enabling dynamic monitoring. Propositions to test include: (1) TLS-positive/PD-L1 low/high bTMB tumors have the highest durable response rate to BCG; (2) TLS-positive/PD-L1 high/low bTMB tumors show graded response consistent with neoantigen availability; (3) noninvasive biomarkers will track with tissue-based immunologic states and predict response earlier than clinical recurrence.

Element	Value
Imaging feature	TLS-like lymphoid aggregates
Anchor gene(s)	CD274, bTMB
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

CD274: Selected as a core mechanistic driver from Tier 1 knowledge graph analysis.

bTMB: bTMB immune-readout axis linking tumor immunogenicity and immune suppression.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	ADNP (CTC mRNA)	Supporting modifier from knowledge graph

Secondary	Urine DNA methylation biomarkers	Supporting modifier from knowledge graph
Secondary	Immunoscore (CD3/CD8 infiltration)	Supporting modifier from knowledge graph
Secondary	MSI	Supporting modifier from knowledge graph
Secondary	TP53	Supporting modifier from knowledge graph
Secondary	ER status	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- ADNP (CTC mRNA): Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- Urine DNA methylation biomarkers: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.